

Outcomes in the Asian subgroup of the phase III randomised HIMALAYA study of tremelimumab plus durvalumab in unresectable hepatocellular carcinoma

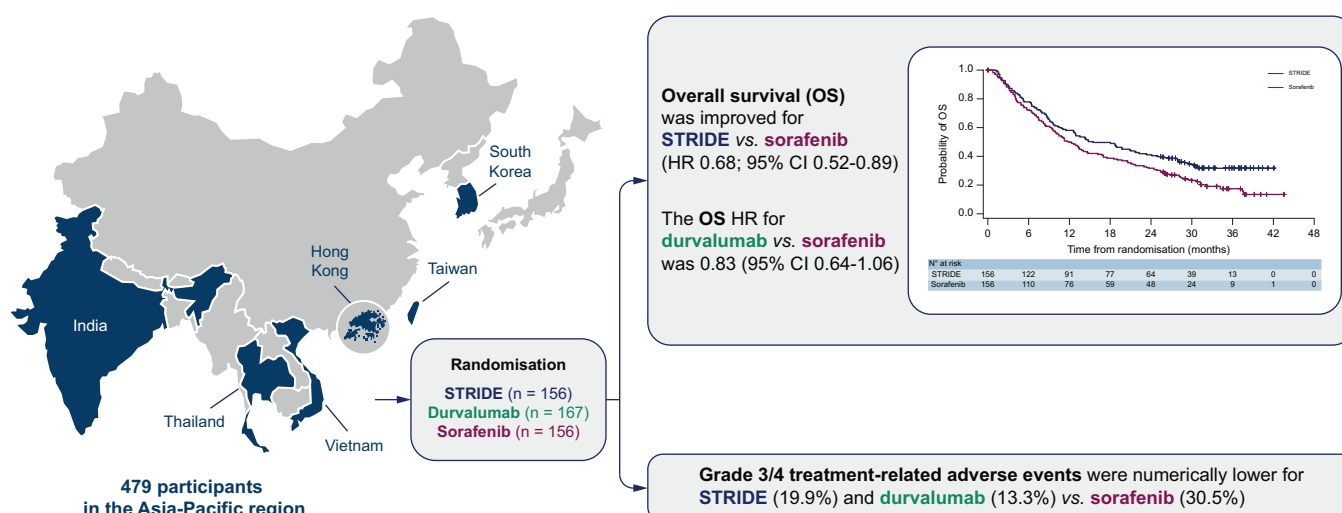
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Graphical abstract



STRIDE improved outcomes versus sorafenib in the Asian subgroup.

These results support the benefits of STRIDE for participants with unresectable hepatocellular carcinoma globally, including the Asia-Pacific region.

Highlights

- The HIMALAYA study showed improved outcomes with STRIDE vs. sorafenib in unresectable HCC.
- In an Asian subgroup of HIMALAYA, STRIDE improved overall survival and objective response rates vs. sorafenib.
- Outcomes were improved with STRIDE vs. sorafenib in the subgroup of participants enrolled in Hong Kong and Taiwan.
- Treatment-related adverse events with STRIDE were generally manageable and low grade in the Asian subgroup.
- STRIDE is beneficial for people with unresectable HCC in the Asia-Pacific region, consistent with the global study population.

Impact and implications

The global, phase III HIMALAYA study found that the STRIDE (Single Tremelimumab Regular Interval Durvalumab) regimen improved overall survival (OS), including long-term OS vs. sorafenib, and that durvalumab monotherapy was non-inferior to sorafenib in participants with unresectable hepatocellular carcinoma (uHCC). However, there are differences in the aetiology and clinical practices related to HCC in parts of Asia, compared to Western countries and Japan, which could lead to differences in treatment outcomes between these regions. The results of this analysis demonstrate the benefits of STRIDE for participants in the Asia-Pacific region, consistent with the full, global study population. Overall, these findings continue to support the use of STRIDE in a diverse population, reflective of uHCC globally.

Outcomes in the Asian subgroup of the phase III randomised HIMALAYA study of tremelimumab plus durvalumab in unresectable hepatocellular carcinoma

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Background & Aims: In the global, phase III HIMALAYA study in unresectable hepatocellular carcinoma (uHCC), STRIDE (Single Tremelimumab Regular Interval Durvalumab) improved overall survival (OS) vs. sorafenib; durvalumab was non-inferior to sorafenib. HBV is the predominant HCC aetiology in most of Asia vs. HCV or non-viral aetiologies in Western countries and Japan. This analysis evaluated safety and efficacy outcomes for STRIDE and durvalumab monotherapy vs. sorafenib, in HIMALAYA participants enrolled in Asia, excluding Japan.

Methods: In HIMALAYA, participants were randomised to STRIDE, durvalumab, or sorafenib. The Asian subgroup in this analysis included participants enrolled in Hong Kong, India, South Korea, Taiwan, Thailand, and Vietnam. OS, objective response rate (ORR; per RECIST, version 1.1), and safety were assessed in the Asian subgroup and in an exploratory subgroup of participants in Hong Kong and Taiwan.

Results: The Asian subgroup included 479 participants randomised to STRIDE (n = 156), durvalumab (n = 167), or sorafenib (n = 156). OS was improved for STRIDE vs. sorafenib (hazard ratio [HR] 0.68; 95% CI 0.52–0.89). The OS HR for durvalumab vs. sorafenib was 0.83 (95% CI 0.64–1.06). In Hong Kong and Taiwan (n = 141), OS HRs for STRIDE vs. sorafenib and durvalumab vs. sorafenib were 0.44 (95% CI 0.26–0.77) and 0.64 (95% CI 0.37–1.08), respectively. In the Asian subgroup, ORR (including unconfirmed responses) was numerically higher for STRIDE (28.2%) and durvalumab (18.6%) vs. sorafenib (9.0%), and Grade 3/4 treatment-related adverse events were numerically lower for STRIDE (19.9%) and durvalumab (13.3%) vs. sorafenib (30.5%).

Conclusions: STRIDE improved outcomes vs. sorafenib in the Asian subgroup. These results support the benefits of STRIDE for participants with uHCC globally, including in the Asia-Pacific region.

Clinical trial number: NCT03298451.

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Introduction

The global, phase III HIMALAYA study evaluated a single dose of tremelimumab, an anti-cytotoxic T-lymphocyte-associated antigen 4 antibody, plus durvalumab, an anti-programmed cell death ligand-1 antibody, in the STRIDE (Single Tremelimumab Regular Interval Durvalumab) regimen and durvalumab monotherapy vs. sorafenib, in participants with unresectable hepatocellular carcinoma (uHCC) who had not been previously treated with systemic therapy.¹ HIMALAYA enrolled a diverse population of participants across Asia, Europe, North America, and South America, reflective of the global uHCC population.¹ The STRIDE

regimen significantly improved overall survival (OS) vs. sorafenib, as demonstrated by an OS hazard ratio (HR) of 0.78 (96.02% CI 0.65–0.93; $p = 0.0035$).¹ OS with durvalumab monotherapy was non-inferior to sorafenib (HR 0.86; 95.67% CI 0.73–1.03; non-inferiority margin, 1.08), but did not demonstrate superiority to sorafenib ($p = 0.0674$).¹ The OS rate at 36 months was 30.7% with STRIDE and 15.1% with sorafenib.¹ An updated OS analysis with 4 years of follow-up continued to support long-term benefit with STRIDE.² Adverse events (AEs) with STRIDE and durvalumab were generally manageable and low grade, and the incidence, frequency, and severity of the AEs were consistent with

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the known safety profiles of tremelimumab and durvalumab.^{1,3} Based on results from HIMALAYA, STRIDE has recently received approval for the treatment of adults with uHCC in many regions across the world, including in the United States, the European Union, Japan, South Korea, and Taiwan.^{4–8}

HCC is the most common type of liver cancer and is one of the leading causes of cancer-related deaths globally.⁹ The highest incidences of HCC globally are observed in East Asia.¹⁰ The estimated age-standardised incidence rates (ASRs) range from 14.3 to 23.0 per 100,000 people in China, South Korea, Thailand, and Vietnam.¹¹ In comparison, ASRs are generally lower in Western countries, with ASRs ranging from 4.3 to 7.6, per 100,000 people in the United States, Canada, the United Kingdom, France, and Germany.¹¹

The aetiology of HCC varies globally.¹⁰ HBV is the predominant aetiology of HCC in most of Asia, accounting for at least 60% of cases, whereas HCV or non-viral aetiologies, such as metabolic dysfunction-associated fatty liver disease (non-alcoholic fatty liver disease and non-alcoholic steatohepatitis), diabetes mellitus, and alcohol-related liver disease, are more common in Japan and Western countries.^{10,12–14} Within recent years, there have been epidemiological shifts in the incidence of HCC from viral-related disease to non-viral-related disease.¹⁵ Studies have shown that the incidence of non-viral aetiologies of HCC is currently on the rise in several countries across the globe.^{12,13}

Compared with Western countries, parts of East Asia have different clinical practices for HCC, such as more intense surveillance protocols.¹⁰ Additionally, studies have shown differences in treatment outcomes throughout Asia. In phase III trials of sorafenib, a trend towards shorter survival was observed in participants enrolled in China, South Korea, and Taiwan in the Asia-Pacific trial relative to those enrolled in predominantly Western countries in the parallel SHARP trial.^{16,17} Furthermore, trends toward better relative outcomes with immune checkpoint inhibitors have been observed for HBV-infected HCC (predominant in parts of Asia) compared with other aetiologies.^{10,18–20}

Given the high HCC burden in Asia and the differences in HCC aetiology in parts of Asia compared to Western countries and Japan, there is a need to understand the effects of STRIDE in this geographic region. This exploratory analysis evaluated the efficacy and safety of STRIDE and durvalumab monotherapy, vs. sorafenib, in participants with uHCC from Asia (excluding Japan) who were enrolled in the HIMALAYA study. An extension of the HIMALAYA study is ongoing for participants enrolled in mainland China, a country which represents a substantial portion of the HCC disease burden worldwide.²¹ As no participants from mainland China were enrolled as part of the global study, an exploratory analysis was performed in participants of Chinese descent in Hong Kong and Taiwan.

Patients and methods

Study design and participants

HIMALAYA (NCT03298451) was a randomised, open-label, multicentre, global, phase III study.¹ The HIMALAYA study design and methodology have been reported in detail previously.¹ Briefly, eligible participants were at least 18 years of age with confirmed uHCC, had received no prior systemic therapy, and were not eligible for locoregional therapy. Participants who had coinfection with HBV and HCV were excluded. Participants

were randomly assigned 1:1:1 to receive STRIDE (tremelimumab 300 mg for one dose plus durvalumab 1,500 mg every 4 weeks), durvalumab (1,500 mg every 4 weeks), or sorafenib (400 mg twice daily).¹ Randomisation was stratified according to macrovascular invasion (yes or no), aetiology of liver disease (HBV or HCV [but not both] or other/non-viral), and ECOG (Eastern Cooperative Oncology Group) performance status (0 or 1).¹

Participants included in the Asian subgroup analysis were enrolled in Hong Kong, India, South Korea, Taiwan, Thailand, and Vietnam. Due to the similarities in the underlying aetiology of HCC between Japan and Western countries,¹⁰ Japan was excluded from the Asian subgroup for the purposes of this analysis. As the mainland China extension cohort of the HIMALAYA study is ongoing, a subgroup analysis was performed on participants enrolled in Hong Kong and Taiwan to assess the outcomes in participants of Chinese descent.

The HIMALAYA study was conducted in accordance with the Declaration of Helsinki and was consistent with the International Council for Harmonisation and Good Clinical Practice guidelines, applicable regulatory requirements, and the Astra-Zeneca Bioethics Policy.¹ The trial protocol was approved by local institutional review boards.¹ Written informed consent was obtained from participants or their legal representatives before participation.¹

Outcomes

We assessed similar outcomes in the Asian subgroup of the HIMALAYA study as those assessed for the global population in the previously published initial analysis (data cut-off: 27 August 2021).¹ Subgroup analyses for OS by region (Asia excluding Japan) in participants treated with STRIDE vs. sorafenib were prespecified for HIMALAYA. OS was defined as the time from date of randomisation until death from any cause. OS rates (at 18, 24, and 36 months) were assessed across treatment arms. Objective response rate (ORR), disease control rate, time-to-response, and duration of response (DoR) per investigator assessment using RECIST version 1.1, were also assessed across treatment arms. Responses did not require confirmation. Safety outcomes were assessed across treatment groups. AEs were assessed using the National Cancer Institute CTCAE version 4.03.¹

Statistical analyses

Efficacy data were analysed in all participants included in the Asian subgroup randomly assigned to STRIDE, durvalumab, and sorafenib. Median OS and landmark OS rates were calculated using the Kaplan-Meier technique. OS HRs were calculated using a stratified Cox proportional hazards model, adjusting for the stratification factors, with a corresponding CI. As this was an exploratory analysis, no alpha adjustment was performed. Safety data were summarised descriptively for all participants included in the Asian subgroup who received at least one dose of study treatment.

Results

Participants

Of 1,171 global participants, 40.9% (n = 479) were enrolled in the Asian region and randomised to STRIDE (n = 156/393), durvalumab (n = 167/389), or sorafenib (n = 156/389).

Demographics and baseline disease characteristics were largely balanced across treatment arms in the Asian subgroup (Table 1). As expected, HBV was the predominant aetiology of HCC in the Asian subgroup, with 62.8% of participants in the STRIDE arm, 58.1% of participants in the durvalumab arm, and 64.1% of participants in the sorafenib arm having HBV. After study treatment discontinuation, 37.8%, 35.9%, and 34.0% of participants in the STRIDE, durvalumab, and sorafenib arms, respectively, received subsequent second-line anticancer therapy (Table S1). Sorafenib was the most common second-line tyrosine kinase inhibitor (TKI) administered in the STRIDE (24.4%) and durvalumab (22.2%) arms, and pembrolizumab was the most common second-line immune checkpoint inhibitor in the sorafenib arm (4.5%). In total, 51.9% of participants in the STRIDE arm, 53.3% of participants in the durvalumab arm, and 43.7% of participants in the sorafenib arm continued treatment beyond disease progression for at least one cycle. Eight participants (5.1%) randomly assigned to STRIDE were retreated with tremelimumab plus durvalumab.

For the subgroup of participants enrolled in Hong Kong and Taiwan, 56 participants were randomised to receive STRIDE, 42 participants were randomised to receive durvalumab, and 43 participants were randomised to receive sorafenib (Table 1). In this subgroup, HBV was the predominant aetiology of HCC, with 76.8%, 66.7%, and 74.4% of participants in the STRIDE, durvalumab, and sorafenib arms, respectively, having HBV. After study treatment discontinuation, 51.8%, 45.2%, and 41.9% of participants in the STRIDE, durvalumab, and sorafenib arms, respectively, received subsequent second-line anticancer therapy (Table S2). Similar to the Asian subgroup, sorafenib was the most common second-line TKI administered in the STRIDE (32.1%) and durvalumab (23.8%) arms, and pembrolizumab was the most common second-line immune checkpoint inhibitor in the

sorafenib arm (11.6%). In total, 100.0% of participants in the STRIDE arm, 100.0% of participants in the durvalumab arm, and 47.8% of participants in the sorafenib arm continued treatment beyond disease progression for at least one cycle. Five participants (15.2%) randomly assigned to STRIDE were retreated with tremelimumab plus durvalumab.

At data cut-off (27 August 2021), the median (range) duration of follow-up in censored participants in the Asian subgroup was 32.2 (25.3–41.9) months with STRIDE, 29.3 (1.9–45.7) months with durvalumab, and 30.0 (0.0–43.6) months with sorafenib. In the subgroup of participants enrolled in Hong Kong and Taiwan, the median (range) duration of follow-up in censored participants was 32.4 (26.5–40.3), 29.9 (27.1–33.7), and 30.0 (0.0–38.0) months for STRIDE, durvalumab, and sorafenib, respectively.

Efficacy

In the Asian subgroup, OS was improved with STRIDE vs. sorafenib (HR 0.68; 95% CI 0.52–0.89; Fig. 1A). Median OS was 16.5 (95% CI 12.6–22.1) months for STRIDE vs. 11.8 (95% CI 9.4–14.7) months for sorafenib. An OS benefit with STRIDE was observed at 36 months, with OS rates of 32.2% for STRIDE vs. 17.3% for sorafenib. In the subgroup of participants enrolled in Hong Kong and Taiwan, the OS HR for STRIDE vs. sorafenib was 0.44 (95% CI 0.26–0.77) (Fig. 2A), and the 36-month OS rate was 49.2% for STRIDE vs. 14.8% for sorafenib.

In the Asian subgroup, the OS HR for durvalumab vs. sorafenib was 0.83 (95% CI 0.64–1.06; Fig. 1B). Median OS was 16.6 (95% CI 12.2–19.2) months for durvalumab vs. 11.8 (95% CI 9.4–14.7) months for sorafenib. The 36-month OS rate was 16.8% for durvalumab vs. 17.3% for sorafenib. For the subgroup of participants enrolled in Hong Kong and Taiwan, the OS HR for durvalumab vs. sorafenib was 0.64 (95% CI 0.37–1.08; Fig. 2B).

Table 1. Participant demographics and baseline characteristics in the Asian subgroup and the Hong Kong and Taiwan subgroup.

Parameter	Asian subgroup			Hong Kong and Taiwan subgroup		
	STRIDE (n = 156)	Durvalumab (n = 167)	Sorafenib (n = 156)	STRIDE (n = 56)	Durvalumab (n = 42)	Sorafenib (n = 43)
Age, median (range), years	60.5 (22–81)	59.0 (24–84)	61.0 (23–88)	62.0 (29–79)	58.5 (29–83)	64.0 (30–88)
Male sex, n (%)	135 (86.5)	138 (82.6)	134 (85.9)	47 (83.9)	37 (88.1)	40 (93.0)
MVI and/or EHS, n (%)	117 (75.0)	123 (73.7)	110 (70.5)	44 (78.6)	26 (61.9)	32 (74.4)
ECOG PS of 0, n (%)	98 (62.8)	98 (58.7)	105 (67.3)	43 (76.8)	28 (66.7)	33 (76.7)
BCLC stage, n (%)						
B	24 (15.4)	27 (16.2)	22 (14.1)	11 (19.6)	13 (31.0)	8 (18.6)
C	132 (84.6)	140 (83.8)	134 (85.9)	45 (80.4)	29 (69.0)	35 (81.4)
ALBI score, n (%)						
1	82 (52.6)	93 (55.7)	91 (58.3)	33 (58.9)	20 (47.6)	20 (46.5)
2	74 (47.4)	74 (44.3)	65 (41.7)	23 (41.1)	22 (52.4)	23 (53.5)
3	0	0	0	0	0	0
Aetiology, n (%)						
HBV*	98 (62.8)	97 (58.1)	100 (64.1)	43 (76.8)	28 (66.7)	32 (74.4)
HCV†	27 (17.3)	31 (18.6)	21 (13.5)	9 (16.1)	9 (21.4)	5 (11.6)
Other/non-viral‡	31 (19.9)	39 (23.4)	35 (22.4)	4 (7.1)	5 (11.9)	6 (14.0)
Race, n (%)						
Asian	155 (99.4)	164 (98.2)	155 (99.4)	56 (100)	42 (100)	53 (100)
Other	1 (0.6)	3 (1.8)	1 (0.6)	0	0	0

Values for the aetiology of liver disease (HBV vs. HCV vs. other/non-viral) and MVI (yes vs. no) were obtained from the pathology at screening.

BCLC, Barcelona Clinic Liver Cancer; ECOG, Eastern Cooperative Oncology Group; EHS, extrahepatic spread; MVI, macrovascular invasion; PS, performance status; STRIDE, Single Tremelimumab Regular Interval Durvalumab.

*Presence of hepatitis B surface antigen and/or anti-hepatitis B core antibody with detectable HBV DNA.

†Presence of anti-HCV antibodies.

‡No active viral hepatitis identified.

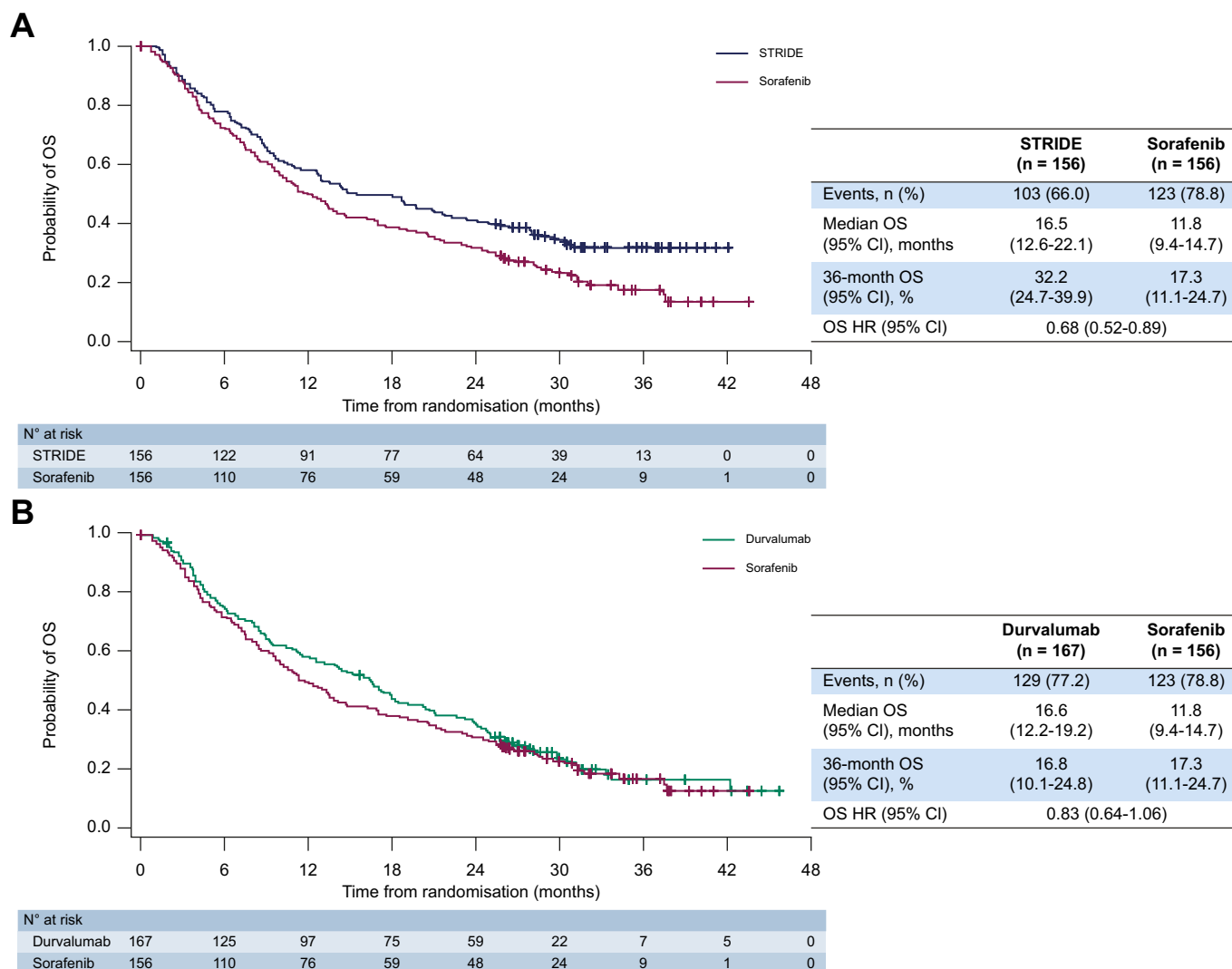


Fig. 1. OS in the Asian subgroup. (A) OS in the Asian subgroup for STRIDE vs. sorafenib and (B) durvalumab vs. sorafenib. Median (range) duration of follow-up in censored participants was 32.2 (25.3-41.9) months with STRIDE, 29.3 (1.9-45.7) months with durvalumab, and 30.0 (0.0-43.6) months with sorafenib. Median OS and landmark OS rates were calculated using the Kaplan-Meier technique. OS HRs were calculated using a stratified Cox proportional hazards model, adjusting for the stratification factors, with a corresponding CI. HR, hazard ratio; OS, overall survival; STRIDE, Single Tremelimumab Regular Interval Durvalumab.

ORR (including unconfirmed responses) was numerically higher for STRIDE (28.2%) and durvalumab (18.6%), vs. sorafenib (9.0%), in the Asian subgroup (Table 2). Median DoR (interquartile range) was 13.7 (4.4-32.2) months for STRIDE, 9.2 (3.7-not reached) months for durvalumab, and 10.2 (4.0-26.0) months for sorafenib, showing that median DoR was numerically longer for STRIDE compared with sorafenib (Table 2). For the subgroup of participants enrolled in Hong Kong and Taiwan, the ORR (including unconfirmed responses) was 33.9% with STRIDE, 23.8% with durvalumab, and 4.7% with sorafenib (Table S3).

Safety

In the Asian subgroup, safety analyses included 472 participants who received at least one dose of STRIDE (n = 156), durvalumab (n = 165), or sorafenib (n = 151). Any AE, regardless of attribution, occurred in 147 (94.2%), 146 (88.5%), and

145 (96.0%) participants receiving STRIDE, durvalumab, and sorafenib, respectively (Table 3). Any treatment-related AEs (TRAEs) occurred in 112 (71.8%) participants receiving STRIDE, 82 (49.7%) participants receiving durvalumab, and 126 (83.4%) participants receiving sorafenib (Table 3). Grade 3/4 TRAEs occurred in 31 (19.9%) participants receiving STRIDE, 22 (13.3%) participants receiving durvalumab, and 46 (30.5%) participants receiving sorafenib. TRAEs leading to discontinuation occurred in 5 (3.2%), 7 (4.2%), and 6 (4.0%) participants receiving STRIDE, durvalumab, and sorafenib, respectively. TRAEs leading to death occurred in 1 (0.6%), 0, and 1 (0.7%) participant receiving STRIDE, durvalumab, and sorafenib, respectively. For the subgroup of participants enrolled in Hong Kong and Taiwan, safety outcomes were generally aligned with safety outcomes in the Asian subgroup (Table S4).

The most common TRAE in participants receiving STRIDE or durvalumab was rash, which occurred in 27.6% and 12.1% of

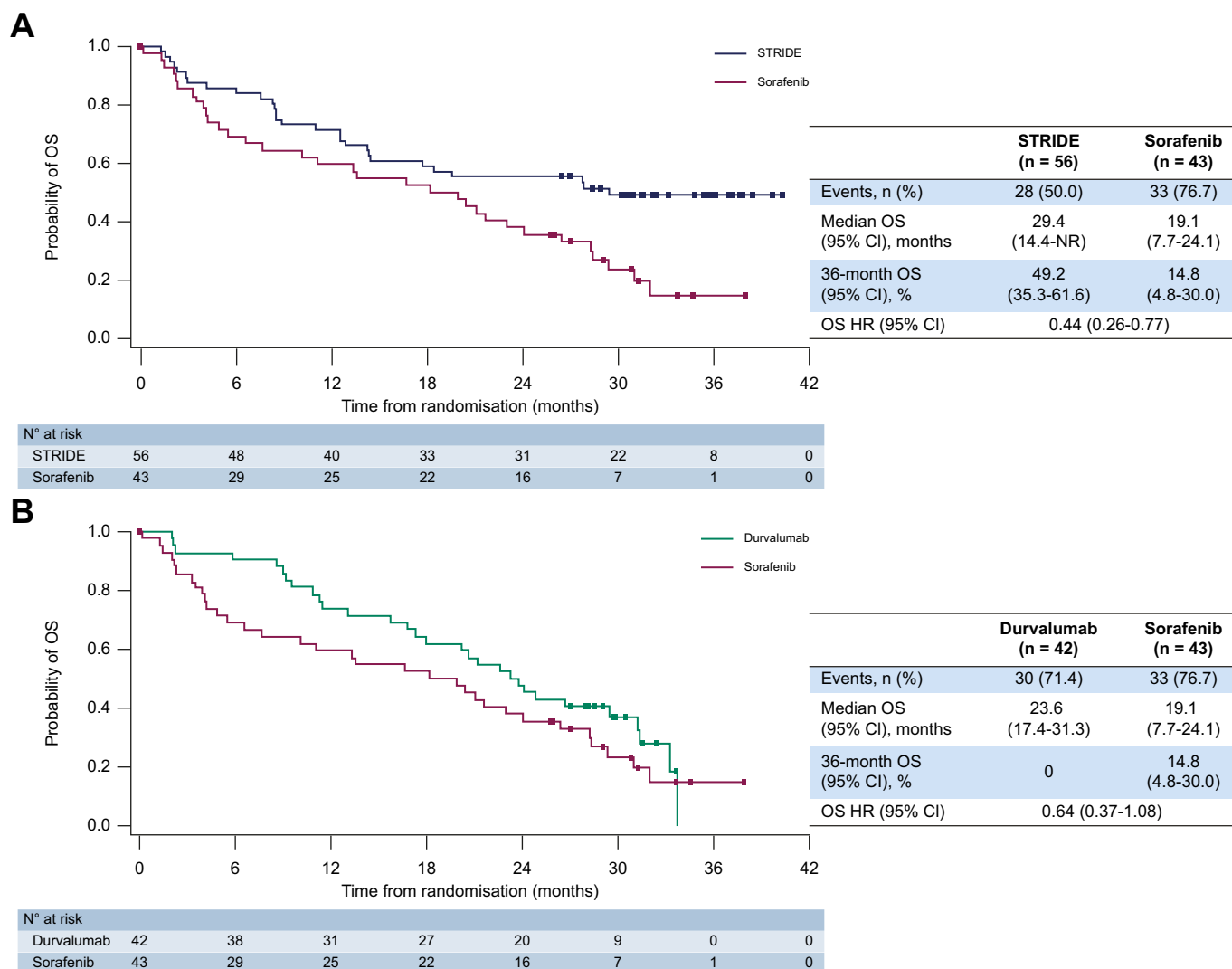


Fig. 2. OS in the Hong Kong and Taiwan subgroup. (A) OS in the Hong Kong and Taiwan subgroup for STRIDE vs. sorafenib, and (B) durvalumab vs. sorafenib. Median (range) duration of follow-up in censored participants was 32.4 (26.5–40.3) months with STRIDE, 29.9 (27.1–33.7) months with durvalumab, and 30.0 (0.0–38.0) months with sorafenib. Median OS and landmark OS rates were calculated using the Kaplan-Meier technique. OS HRs were calculated using a stratified Cox proportional hazards model, adjusting for the stratification factors, with a corresponding CI. HR, hazard ratio; NR, not reached; OS, overall survival; STRIDE, Single Tremelimumab Regular Interval Durvalumab.

participants, respectively (Fig. 3). The most common TRAE in participants receiving sorafenib was palmar-plantar erythrodysesthesia syndrome, which occurred in 58.9% of participants. Any immune-mediated AEs (imAEs) occurred in 47 (30.1%) participants receiving STRIDE, 24 (14.5%) participants receiving durvalumab, and 8 (5.3%) participants receiving

sorafenib (Table 3). The most common imAEs among participants receiving STRIDE were hypothyroidism (9.6%), rash (5.1%), and hyperthyroidism (4.5%; Fig. 4). imAEs associated with liver toxicity, including hepatitis, increased aspartate aminotransferase, or increased alanine aminotransferase, occurred in 1.2%, 0.6%, and 0.6% of participants who received

Table 2. Tumour response (including unconfirmed responses) in the Asian subgroup based on investigator assessment according to RECIST version 1.1.

Parameter	STRIDE (n = 156)	Durvalumab (n = 167)	Sorafenib (n = 156)
ORR, n (%)	44 (28.2)	31 (18.6)	14 (9.0)
CR, n (%)	4 (2.6)	3 (1.8)	0
PR, n (%)	40 (25.6)	28 (16.8)	14 (9.0)
SD, n (%)	48 (30.8)	59 (35.3)	69 (44.2)
DCR, n (%)	92 (59.0)	90 (53.9)	83 (53.2)
Median TTR (IQR),* months	2.1 (1.8-3.9)	1.9 (1.8-3.7)	3.7 (1.9-3.9)
Median DoR (IQR),* months	13.7 (4.4-32.2)	9.2 (3.7-NR)	10.2 (4.0-26.0)

CR, complete response; DCR, disease control rate; DoR, duration of response; IQR, interquartile range; NR, not reached; ORR, objective response rate; PR, partial response; SD, stable disease; STRIDE, Single Tremelimumab Regular Interval Durvalumab; TTR, time-to-response.

*In participants with a response.

Table 3. Safety outcomes in the Asian subgroup.

Participants with event, n (%)	STRIDE (n = 156)	Durvalumab (n = 165)	Sorafenib (n = 151)
AE	147 (94.2)	146 (88.5)	145 (96.0)
Grade 3/4 AE	73 (46.8)	56 (33.9)	70 (46.4)
TRAE	112 (71.8)	82 (49.7)	126 (83.4)
Grade 3/4 TRAE	31 (19.9)	22 (13.3)	46 (30.5)
TRAE leading to discontinuation	5 (3.2)	7 (4.2)	6 (4.0)
TRAE leading to death	1 (0.6)	0	1 (0.7)
Any imAE	47 (30.1)	24 (14.5)	8 (5.3)

Treatment causality was assessed by the investigator.

Includes AEs with an onset date on or after the date of first dose or pre-treatment AEs that increase in severity on or after the date of first dose up to and including 90 days following the date of last dose of study medication or up to and including the date of initiation of the first subsequent therapy (whichever occurs first).

AE, adverse event; imAE, immune-mediated adverse event; STRIDE, Single Tremelimumab Regular Interval Durvalumab; TRAE, treatment-related adverse event.

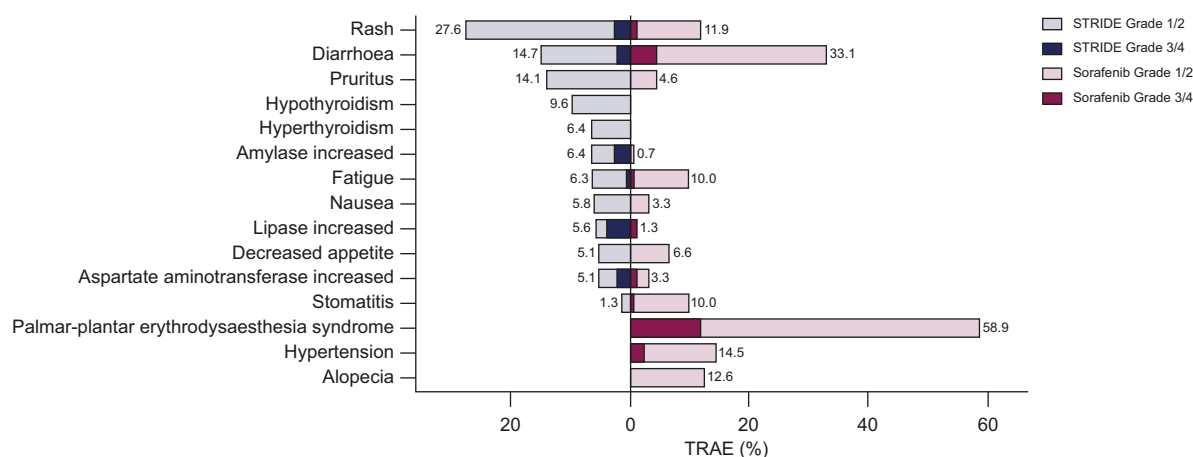
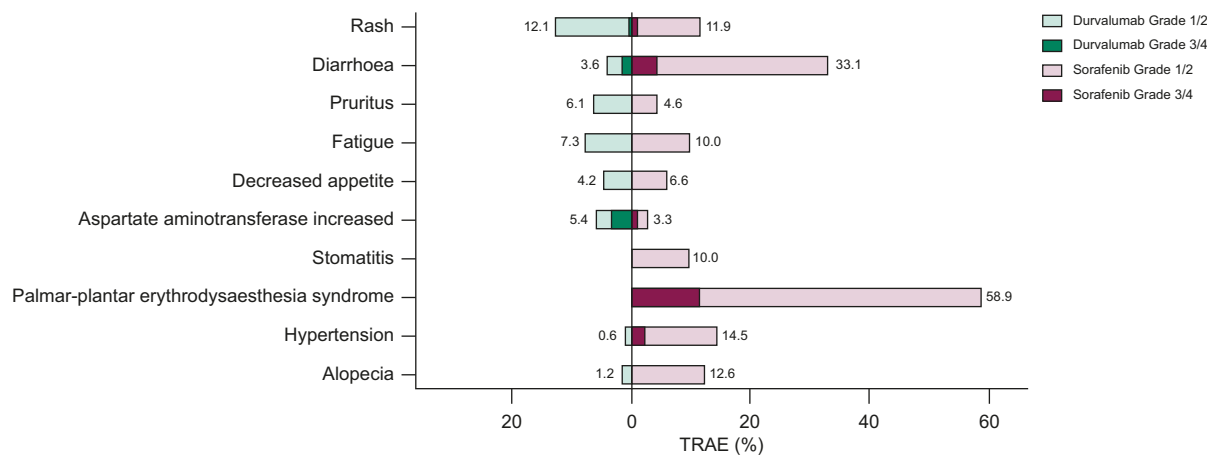
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Fig. 3. Most common TRAEs in the Asian subgroup. (A) Most common TRAEs ($\geq 5\%$ in any treatment arm) in the Asian subgroup for STRIDE vs. sorafenib and (B) durvalumab vs. sorafenib. Treatment causality was assessed by the investigator. Includes AEs with an onset date on or after the date of first dose or pre-treatment AEs that increase in severity on or after the date of first dose up to and including 90 days following the date of last dose of study medication or up to and including the date of initiation of the first subsequent therapy (whichever occurs first). AE, adverse event; STRIDE, Single Tremelimumab Regular Interval Durvalumab; TRAE, treatment-related adverse event.

STRIDE, and 1.2%, 3.6%, and 3.0% of participants who received durvalumab, respectively (Fig. 4).

Discussion

The HIMALAYA study was a large, robust trial stratified by aetiology of HCC that enrolled a diverse, global population. The

trial showed a benefit for participants treated with STRIDE compared with sorafenib, across clinically relevant subgroups.^{1,2} Findings from this exploratory analysis of the Asian subgroup of participants in HIMALAYA continue to support the efficacy of STRIDE across geographic regions.

In this analysis, STRIDE improved efficacy outcomes vs. sorafenib, consistent with the global population.¹ OS was

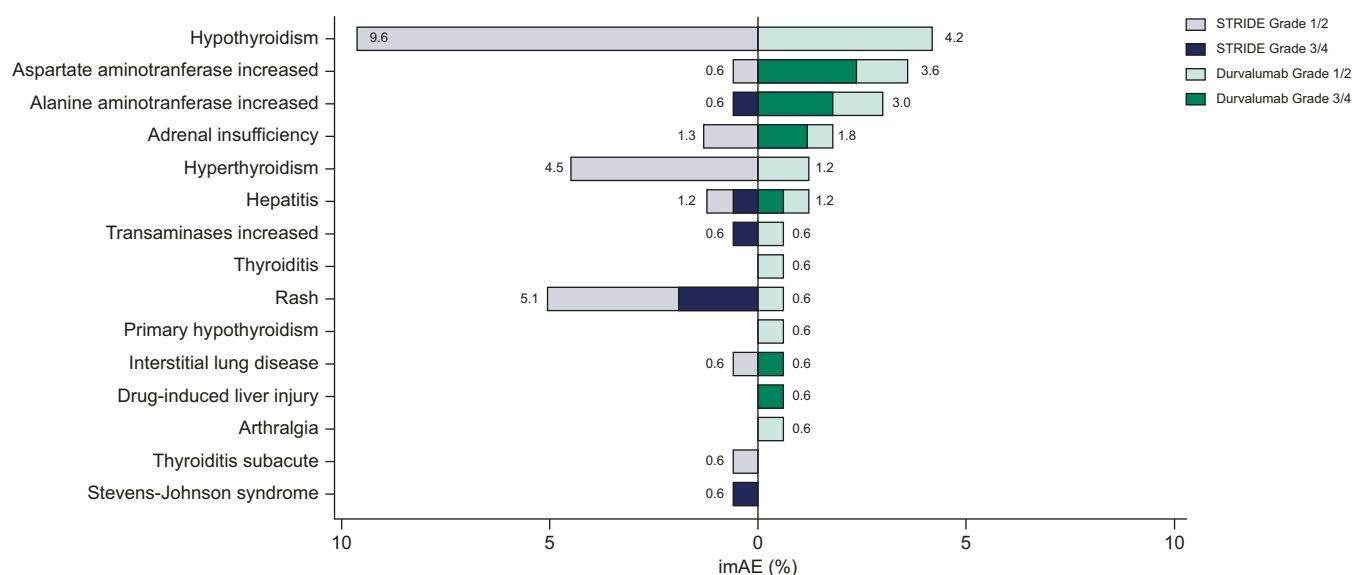


Fig. 4. imAEs in the Asian subgroup. imAEs in the Asian subgroup for STRIDE vs. durvalumab. Includes AEs with an onset date on or after the date of first dose or pre-treatment AEs that increase in severity on or after the date of first dose up to and including 90 days following the date of last dose of study medication or up to and including the date of initiation of the first subsequent therapy (whichever occurs first). AE, adverse event; imAE, immune-mediated adverse event; STRIDE, Single Tremelimumab Regular Interval Durvalumab.

improved for STRIDE vs. sorafenib in the Asian subgroup. Furthermore, STRIDE demonstrated an approximately 1 in 3 OS rate at 36 months (32.2%), which was higher than with sorafenib (17.3%). ORR was numerically higher for STRIDE vs. sorafenib in the Asian subgroup, and DoR was numerically longer for STRIDE. STRIDE also demonstrated improved efficacy outcomes vs. sorafenib in the subgroup of participants enrolled in Hong Kong and Taiwan. OS with durvalumab vs. sorafenib in the Asian subgroup, and in Hong Kong and Taiwan, were consistent with the statistical non-inferiority observed for the global population,¹ further supporting the clinical benefit of combining tremelimumab with durvalumab over durvalumab monotherapy in these geographic regions.

Participants enrolled in the Asian subgroup made up 40.9% of the global population. Baseline demographics, disease characteristics, and frequency of subsequent therapy in the Asian subgroup were generally consistent with the global population,¹ with the exception of a younger median age (STRIDE: 60.5 vs. 65.0 years; durvalumab: 59.0 vs. 64.0 years; sorafenib: 61.0 vs. 64.0 years) and a higher frequency of HBV across treatment arms (STRIDE: 62.8% vs. 31.0%; durvalumab: 58.1% vs. 30.6%; sorafenib: 64.1% vs. 30.6%), which is aligned with established trends of regional aetiological variation.¹⁰ ORRs were numerically higher for the STRIDE arm in the Asian subgroup (28.2%) than in the global population (20.1%). The median DoR was numerically longer with STRIDE vs. sorafenib in the Asian subgroup (13.7 vs. 10.2 months), and numerically shorter for STRIDE compared with the global population (13.7 vs. 22.3 months).¹ Median OS with STRIDE vs. sorafenib was similar to the global population (STRIDE: 16.5 vs. 16.4 months; sorafenib: 11.8 vs. 13.8 months), with similar OS rates at 36 months reported for STRIDE (32.2% vs. 30.7%), and slightly lower OS rates at 36 months reported for sorafenib (17.3% vs. 20.2%).¹ The lower OS rates at 36 months reported for sorafenib in the Asian subgroup compared with the global population aligns with the high prevalence of HBV-infected

participants in the Asian region and with literature describing the development of sorafenib resistance in HBV-infected HCC.^{10,22} Overall, a greater numerical improvement in OS with STRIDE vs. sorafenib was observed in the Asian subgroup (HR 0.68; 95% CI 0.52-0.89) than in the global population (HR 0.78; 95% CI 0.66-0.92), although no formal statistical comparison was performed.

In the subgroup of participants enrolled in Hong Kong and Taiwan, baseline demographics and disease characteristics were generally consistent with the global population, with the exception of younger median age in the STRIDE and durvalumab arms (STRIDE: 62.0 vs. 65.0 years; durvalumab: 58.5 vs. 64.0 years), a higher frequency of HBV across treatment arms (STRIDE: 76.8% vs. 31.0%; durvalumab: 66.7% vs. 30.6%; sorafenib: 74.4% vs. 30.6%), and more participants with an ECOG performance status of 0 (STRIDE: 76.8% vs. 62.1%; durvalumab: 66.7% vs. 60.9%; sorafenib: 76.7% vs. 62.0%).¹ After study treatment discontinuation, rates of subsequent therapy use were generally similar compared with the global population, with a difference of $\geq 10.0\%$ noted in the STRIDE arm (STRIDE: 51.8% vs. 40.7%; durvalumab: 45.2% vs. 43.2%; sorafenib: 41.9% vs. 45.0%).¹ In addition, more participants continued treatment beyond disease progression for at least one cycle in Hong Kong and Taiwan when compared with the global population (STRIDE: 100.0% vs. 46.9%; durvalumab: 100.0% vs. 48.5%; sorafenib: 47.8% vs. 35.8%).¹ Median OS was improved with STRIDE vs. sorafenib in participants enrolled in Hong Kong and Taiwan, and was longer compared with the global population (STRIDE: 29.4 vs. 16.4 months; sorafenib: 19.1 vs. 13.8 months).¹ Moreover, higher OS rates were reported at 36 months for STRIDE vs. sorafenib when compared with the global population (STRIDE: 49.2% vs. 30.7%; sorafenib: 14.8% vs. 20.2%), with a greater numerical improvement in OS HRs (0.44 [95% CI 0.26-0.77] vs. 0.78 [95% CI 0.65-0.93]).¹ The ORR was also substantially higher for the STRIDE arm in the subgroup of participants enrolled in Hong

Kong and Taiwan (33.9%) than the Asian subgroup (28.2%) or the global population (20.1%).¹ These participants enrolled in Hong Kong and Taiwan may represent people with uHCC of Chinese descent, with China accounting for a large proportion of the liver cancer burden worldwide.²¹

While results from the HIMALAYA study support the efficacy of STRIDE across the global study population, the trend toward further improvement in OS HRs in the Asian region with immunotherapy-based regimens is consistent with results reported from other phase III studies of immune checkpoint inhibitors in uHCC, and may be associated with aetiology or other covariates associated with the Asian region.^{23–25} Studies have reported that immune checkpoint inhibitors have favourable outcomes in people with HBV-related uHCC.^{18–20,25} Moreover, studies have shown that sorafenib may perform worse in people with HBV-related HCC, compared with other aetiologies,^{16,26} and evidence indicates that HBV infection can induce resistance to sorafenib,²⁷ which may negatively impact the efficacy of sorafenib in Asian regions. In HIMALAYA, a numerically shorter median OS with sorafenib was observed in the Asian subgroup (11.8 months) compared to the global population (13.8 months).¹ For the subgroup of participants located in Hong Kong and Taiwan, where the percentage of participants with HBV-related uHCC was higher than that of the rest of the Asian region, the higher median OS with STRIDE (29.4 months) contributed to a lower OS HR, despite a numerically higher median OS with sorafenib (19.1 months) compared with the global population.¹

Overall, the consistent cross-trial trends for improved outcomes with immune checkpoint inhibitors in parts of Asia compared with the global population^{23,25,28} and, in particular, for those with HBV-related HCC,^{18–20,25,28} suggests that outcomes for Asian subgroups may influence the overall results of global trials. While the HIMALAYA study did not enrol participants from mainland China as part of the global study, the global cohort of the IMbrave150 study was comprised of a substantial proportion of participants from mainland China who were predominantly HBV-infected.^{1,23,29} The OS and response data from this exploratory analysis in both the Asian and Hong Kong and Taiwan subgroups of HIMALAYA, aligns with that observed in the global and Chinese cohorts of the IMbrave150 study.^{23,29} Overall, these outcomes support the improved clinical benefit of immunotherapy-containing regimens, in particular, when compared with sorafenib, including in subsets

of Asian patients and in those who are HBV positive. Therefore, overall outcomes for global studies are likely influenced by the relative representation of geographic subgroups, posing limitations for the interpretation of cross-trial comparisons.

Similar to the global population, safety profiles were tolerable and manageable for both STRIDE and durvalumab monotherapy in the Asian subgroup and were differentiated from other current uHCC therapies.^{1,16,29,30} Overall, safety outcomes in the Asian subgroup and in participants enrolled in Hong Kong and Taiwan were generally consistent with the global population, reporting similar rates of TRAEs, Grade 3/4 TRAEs, TRAEs leading to discontinuation, and TRAEs leading to death.¹ Previous findings have shown that acute liver toxicities may occur after immune activation in HBV, and are rarely seen with HCV.^{31,32} In this analysis, no safety concerns specific to the Asian subgroup were identified, and the numerically improved efficacy of STRIDE in the Asian subgroup and subgroup of participants enrolled in Hong Kong and Taiwan was not associated with any apparent difference in the safety profiles in these regions. These findings are important given the high proportion of HBV aetiology in these subgroups and suggest that liver toxicity is not a concern with STRIDE. Taken together, safety results support a tolerable and manageable safety profile with STRIDE in the Asian subgroup and subgroup of participants enrolled in Hong Kong and Taiwan, consistent with the global population.¹

This study has some limitations. The subgroup analysis was hypothesis generating and was not powered for statistical comparison; no type 1 error adjustment was made. Additionally, while participants from Hong Kong and Taiwan may reflect outcomes for people of Chinese descent, a definitive analysis of participants from mainland China has not yet been done. Furthermore, there is heterogeneity in the population of people with uHCC across Asia,¹⁰ and results of this analysis may not be fully representative of each individual country or region included in the Asian subgroup. Lastly, the proportion of participants in the Asian subgroup who had radiologic or histologic cirrhosis was not documented; consequently, whether the presence of cirrhosis influenced outcomes in the Asian subgroup was not determined.

In conclusion, these results support the benefit of STRIDE for participants with uHCC in the Asia-Pacific region and continue to support the positive benefit-risk profile of this regimen in a diverse population, reflective of uHCC globally.

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Abbreviations

AE, adverse event; ASR, age-standardised incidence rate; DoR, duration of response; HCC, hepatocellular carcinoma; HR, hazard ratio; imAE, immune-mediated adverse event; ORR, objective response rate; OS, overall survival; STRIDE, Single Tremelimumab Regular Interval Durvalumab; TRAE, treatment-related adverse event; uHCC, unresectable hepatocellular carcinoma.

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Conflicts of interest

George Lau reports receiving consultant fees from AstraZeneca. Ghassan K. Abou-Alfa reports receiving grant and research support from Arcus, AstraZeneca, BioNTech, Bristol Myers Squibb, Celgene, Flatiron, Genentech/Roche, Genoscience, Incyte, Polaris, Puma, QED, Silenseed, and Yiviva; reports consultant fees from Adicet, Alnylam, AstraZeneca, Autem, Beigene, Berry Genomics, Boehringer Ingelheim, Celgene, Cend, CytomX, Eisai, Eli Lilly, Exelixis, Flatiron, Genentech/Roche, Genoscience, Helio, Helsinn, Incyte, Ipsen, Merck, Nerviano, Newbridge, Novartis, QED, Redhill, Rafael, Servier, Silenseed, Sobi, Vector, and Yiviva; and reports filed patent PCT/US2014/031545, filed: 24 March 2014, and priority application Serial No.: 61/804,907, filed: 25 March 2013. Ann-Li Cheng reports receiving advisory board fees from AstraZeneca, Bayer, BeiGene, Bristol Myers Squibb, Exelixis, F. Hoffmann-La Roche, Genentech, Ipsen, and Merck; reports being an invited speaker for Amgen, Bayer, Chugai Pharma, Eisai, Novartis, and Ono Pharmaceutical; and reports receiving travel grants from IQVIA. Wattana Sukeepaisarnjaroen has nothing to disclose. Tu Van Dao reports receiving consultant fees from AstraZeneca, Bayer, Eisai, Ipsen, MSD, Novartis, Pfizer, Pierre Faber, Roche, and Taiho Pharmaceutical. Yoon Koo Kang reports receiving advisory board fees from ALX Oncology, Amgen, Blueprint, Bristol Myers Squibb, Daehwa, MacroGenics, Merck, Novartis, Roche, Surface Oncology, and Zymeworks. Sathesh Chiradoni Thungappa reports receiving grant or research support from AstraZeneca and Eisai; and consultant fees from AstraZeneca. Masatoshi Kudo was an invited speaker for Bayer, Chugai Pharma, Eisai, Eli Lilly, MSD, and Takeda; and reports receiving research funding (to institution) from AbbVie, Chugai Pharma, EA Pharma, Eisai, GE Healthcare, Gilead Sciences, Ono Pharmaceutical, Otsuka, Sumitomo Dainippon Pharma, Taiho Pharmaceutical, and Takeda. Bruno Sangro reports receiving advisory board fees from AstraZeneca, Bayer, Bristol Myers Squibb, Boston Scientific, Eisai, Incyte, Ipsen, Roche, Sirtex, and Terumo; reports being an invited speaker for AstraZeneca, Bayer, Bristol Myers Squibb, Eisai, Incyte, Ipsen, Roche, and Sirtex; and reports receiving research funding (to institution) from Bristol Myers Squibb and Sirtex. Robin Kate Kelley reports receiving consultant or advisory board fees (to institution) from Agios, AstraZeneca, Exelixis, Ipsen, and MSD, and (to self) from Exact Sciences, GlaxoSmithKline, Ipsen, J Pharma, Kinnate, Moderna, Regeneron, and Tyra Biosciences; and reports receiving research funding from Adaptimmune, Agios, AstraZeneca, Bayer, Bristol Myers Squibb, Eli Lilly, EMD Serono, Exelixis, Genentech, Loxo Oncology, MSD, Novartis, Partner Therapeutics, QED, Relay Therapeutics, Roche, Surface Oncology, and Taiho Pharmaceutical (all to institution). Junji Furuse reports receiving grant or research support from Astellas, Chugai Pharma, Daiichi Sankyo, Eisai, Incyte Japan, J-Pharma, Merck Bio, Mochida, MSD, Ono Pharmaceutical, Taiho Pharmaceutical, Takeda, Sanofi, Sumitomo Dainippon Bayer, and Yakult Honsha; and consultant fees from Bayer, Chugai Pharma, Daiichi Sankyo, EA Pharma, Eisai, Eli Lilly Japan, Incyte Japan, Kyowa Hakko Kirin, MSD, Mylan EPD, Novartis, Ono Pharmaceutical, Pfizer, Sanofi, Servier Japan, Taiho Pharmaceutical, Takeda, Teijin Pharma, and Yakult Honsha. Joong-Won Park reports receiving consulting or advisory fees from AstraZeneca, BeiGene, and Roche; reports honoraria fees from AstraZeneca, Bayer, Eisai, and Roche; and reports research funding from AstraZeneca, Bristol Myers Squibb, Eisai, Exelixis, MSD, Ono Pharmaceutical, and Roche. Patrapim Sunpaweravong reports receiving advisory board or speaker fees from Amgen, AstraZeneca, Bayer, Bristol Myers Squibb, Daiichi Sankyo, Eisai, Ipsen, Janssen, Juniper Biologics, Mundipharma, MSD, Novartis, Pfizer, Roche, Taiho Pharmaceutical, and ZP Therapeutics; reports research funding (to institution) from Amgen, AstraZeneca, Exscientia, Janssen, Mirati Therapeutics, MSD, Novartis, and Roche; and reports travel grants from Amgen, AstraZeneca, Bayer, Bristol Myers Squibb, DKSH, Eisai, Janssen, Juniper Biologics, MSD, Novartis, Pfizer, Roche, Sanofi, Taiho Pharmaceutical, and ZP Therapeutics. Angelica Fasolo reports receiving research funding (to institution) from Roche and Bristol Myers Squibb. Thomas Yau reports receiving advisory board fees from AstraZeneca, Bristol Myers Squibb, MSD, and Roche. Tomokazu Kawaoka reports receiving research funding (to institution) from AstraZeneca. Sergio Azevedo reports receiving grant and research support from Amgen, AstraZeneca, Bristol Myers Squibb, MSD, Novartis, Pfizer, and Roche/

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Please refer to the accompanying ICMJE disclosure forms for further details.

Authors' contributions

GKA-A and AN contributed to the conception and design of the study. GL, GKA-A, MK, J-WP, PS, AF, SA, and EA contributed to data collection. GL, GKA-A, A-LC, WS, TVD, YKK, SCT, MK, BS, RKK, JF, J-WP, AF, TY, TK, SA, MR, EA, MY, ARH, MM, CG, AN, and SLC contributed to data analysis and interpretation. GKA-A, MK, and AF contributed to study supervision. All authors contributed to the writing of the manuscript and approved the final version.

Data availability statement

Data underlying the findings described in this manuscript may be obtained in accordance with AstraZeneca's data sharing policy described at: <https://astrazenecagrouptrials.pharmacm.com/ST/Submission/Disclosure>. Data for studies directly listed on Vivli can be requested through Vivli at www.vivli.org. Data for studies not listed on Vivli could be requested through Vivli at <https://vivli.org/members/enquiries-about-studies-not-listed-on-the-vivli-platform/>. AstraZeneca Vivli member page is also available outlining further details: <https://vivli.org/ourmember/astrazeneca/>.

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Supplementary data

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Author names in bold designate shared co-first authorship

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